

**Information
Visualization**

Data sense making (2)

Lesson 6

Marilena Daquino
Assistant Professor

Department of
Classical Philology
and Italian Studies

marilena.daquino2@unibo.it

Table of contents

01 Other selected tasks

Geo-spatial, topical, network
analysis

02 Hands-on

Data analysis with python



01

Selected tasks

Maps, flows, and graphs

Statistics

Profiling (micro-meso-macro level)

Types of analysis

Temporal

WHEN: evolution of variables over time

Geo-spatial

WHERE: trajectories and space dimension of variables

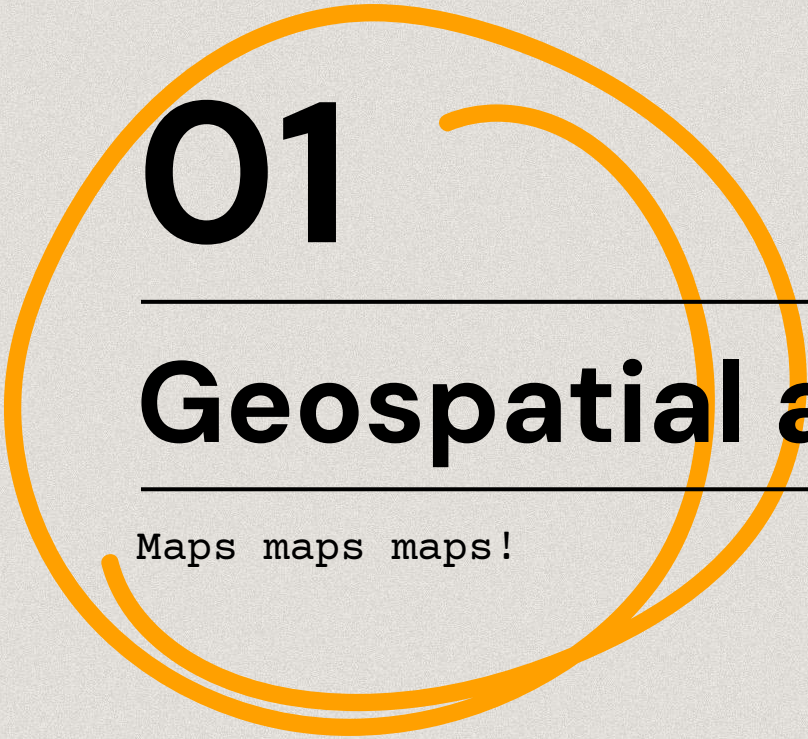
Topical

WHAT: analysis of categorical variables

Network

WITH WHOM: relations and distance between data points

Types of analysis

A large, hand-drawn style orange circle with a thick stroke, partially enclosing the text '01' and 'Geospatial analysis'.

01

Geospatial analysis

Maps maps maps!

Purpose

We see
this
today!

[illegible]

The advance (tan) and disastrous retreat (black) of Napoleon's Grande Armee through Russia.

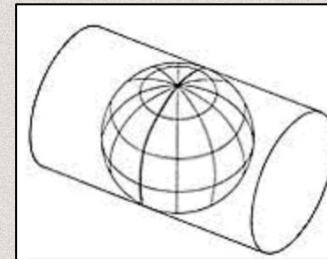
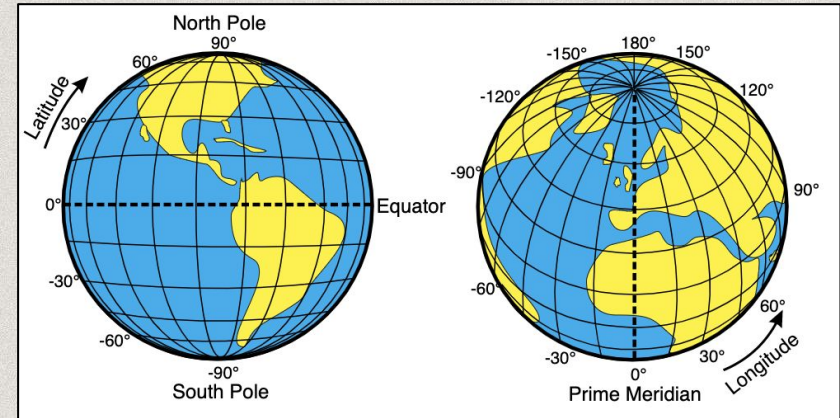
Geographic maps

01

Geospatial analysis

Fundamentals

Data must be accompanied by geospatial information, such as **latitude and longitude** used to position a **shape** on an abstraction of the earth, based on a chosen **projection** system.



Universal Transverse Mercator (UTM) Grid.

The equator is used as reference point for defining North and South.

Geographic maps

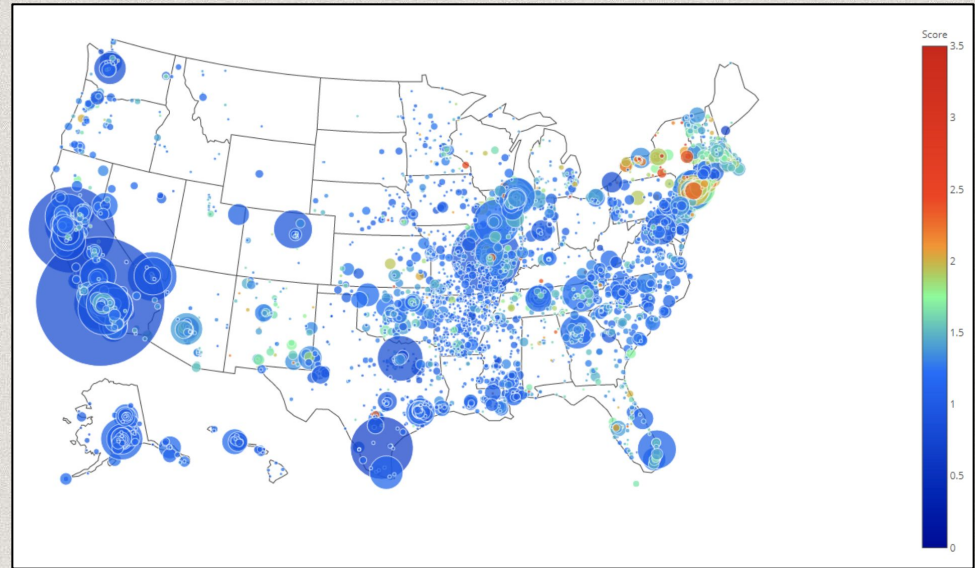
Geospatial analysis

Fundamentals

A shape can be a **point** (or a glyph), identified by two coordinates placed on cartesian axes

x=long; y=lat

Other variables can be represented by the size (numeric), the color (categorical), or the gradation (ordinal).



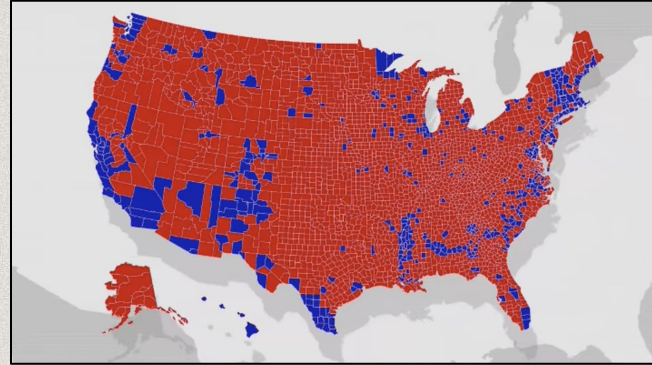
Geographic maps

Fundamentals

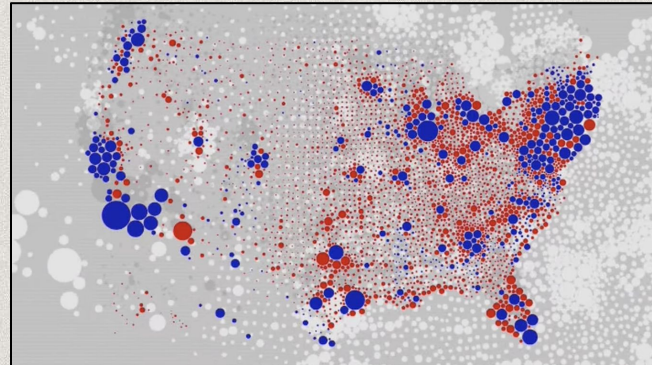
A shape can be an **area**, colored according to categorical and numeric values (e.g. US election votes).

However, areas can be misleading, since larger areas are associated to high numbers (but it might not be the case).

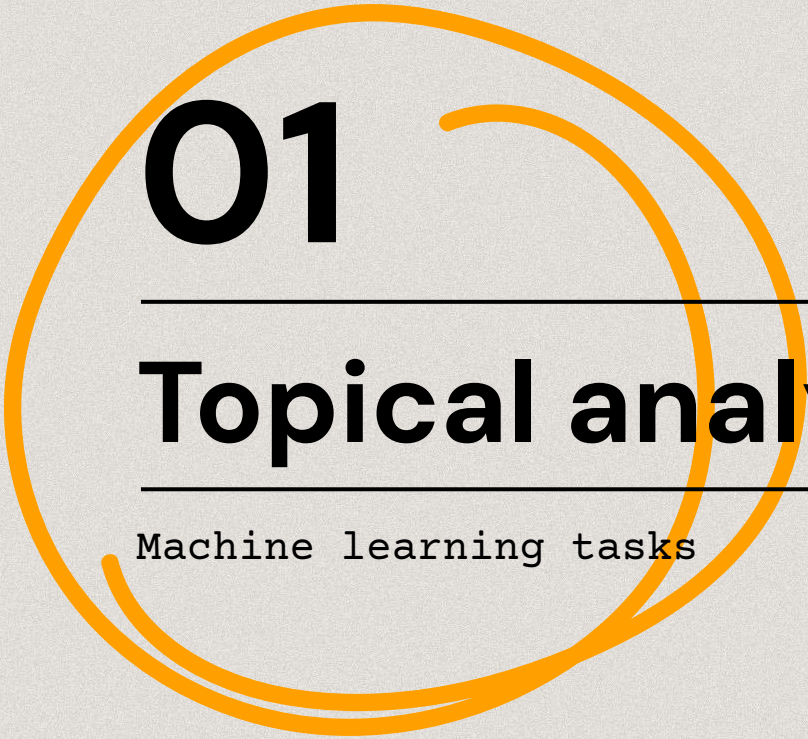
Geospatial analysis



2020 US elections, countries by winning party.



2020 US elections, countries by number of voters.



01

Topical analysis

Machine learning tasks

Topical analysis

01

Topical analysis

Purpose

Analyse the **occurrences** of a topic in texts, images, or structured data to understand distribution, dependencies with other topics, correlation, etc.

Typical tasks include **classification** (e.g. of documents, images), **recommendation** (e.g. similar docs), and **prediction** (e.g. new docs).

Topical analysis is often used to refer to news and social media analysis, where occurrences of a topic in media is analysed to understand a phenomenon (e.g. propaganda, fake news, terrorism).

Topical analysis

Topical analysis

Methods

Mainly based on Machine Learning algorithms

- **Texts.** Natural Language Processing
- **Images.** Computer vision
- **Structured data.** Other probabilistic and rule-based models

Natural Language Processing

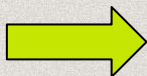
01

Topical analysis

Basic tasks

- **Tokenization.** Extraction of single words from sentences.
- **Lemmatization.** Extraction of lemmas from words
- **Part Of Speech (POS) tagging.** Annotation of linguistic and morphological elements in sentences.
- **Named Entity Recognition (NER).** Identify strings that represent real-world entities (people, organisations, places, dates, etc.)
- **Co-referencing and relation extraction.** Identify named entities not explicitly mentioned, and their relation.

*We see
this
in the
next
class!*

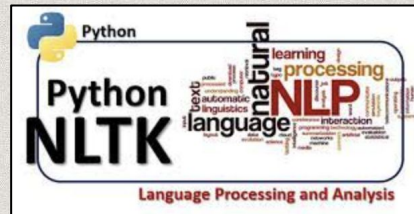


Natural Language Processing

More sophisticated tasks

- **Text classification.** Annotate documents with categories. Methods include topic modelling.
- **Sentiment analysis.** Categorize texts based on the valence of sentiment (positive, negative, neutral)
- **Text summarisation.** Reduces the length of one or more texts to smaller, meaningful, summaries.
- **Question answering.** Retrieve information based on a natural language question.
- **Text generation.** Generate texts from image/text inputs.
- And more...

Topical analysis

The logo for spaCy, a natural language processing library. It features the word "spaCy" in a white, sans-serif font on a blue rectangular background.

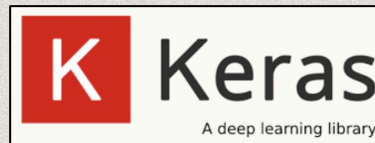
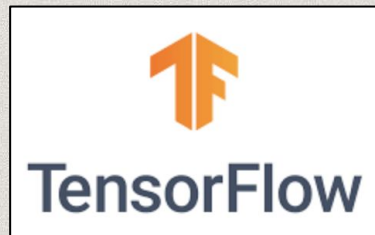
Computer vision

01

Topical analysis

General tasks

- **Object detection.** Identify the presence and the contour of objects in an image.
- **Image classification.** Associate categories to images, based on their subjects.
- **Object recognition.** Identify a specific instance of a class (e.g. a person, a building)
- **Image restoration.** Reconstruction of original color palette.



Machine Learning

01

Topical analysis

General tasks

- **Classification.** Associate a class or category to texts, images, data.
- **Recommendation.** Based on classification, provides a ranking of data based on categories.
- **Prediction.** Estimate next values in a series of continuous values (e.g. prices of artworks next month)

Machine Learning

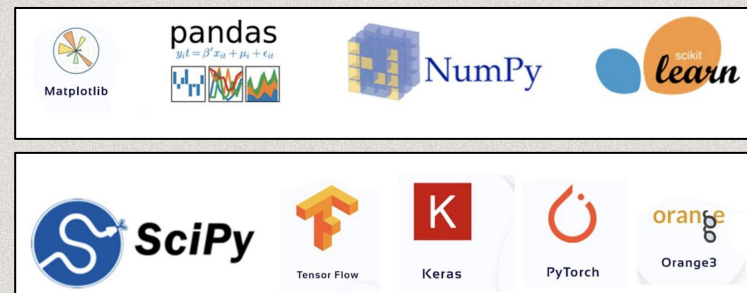
01

Topical analysis

Some tasks

- **Clustering.** Highlight groupings, e.g. identify groups of people w/ a common behaviour.
- **Outlier detection.** Extract anomalies, e.g. online frauds. Looks for differences rather than similarities.
- **Association rule mining.** Extract patterns that identify groups, e.g. products frequently bought together.

We see
this
today!



If you are interested, have a look at this [tutorial](#)

Apriori algorithm

Topical analysis

What

Look for groups of entities that are **likely** to occur together

What input

Groups of co-occurring items in the form of lists (of different lengths)

What is not

It does not look for similarities or differences, but for **correlation**

What output

Groups of co-occurring items in the form of lists, called **rules**, ranked by **likelihood**

Apriori algorithm

An algorithm to find association rules.

It tries all the combinations of items (w/ a minimum frequency).

It generates a rule representing the **probability** of co-occurrence.

Apriori algorithm

Topical analysis

Apriori algorithm

We have 5 customers in a supermarket, and we know what is in their basket.

What are the items that are mostly bought together?

<i>TID</i>	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

Apriori algorithm

01

Topical analysis

Rules

Rules are in the form

IF {Antecedent} THEN {Consequent}

A -> C

Caveat

Even a small dataset creates a huge number of association rules, which have to be interpreted and pruned.

Pruned rules include trivial rules or/and inexplicable rules.

You must see this!

Measures

SUPPORT. Measures how frequently items of the rule occur together in the dataset.

CONFIDENCE. Measures the conditional **probability** that the consequent will occur when there is the antecedent.

Rules with high support and confidence are interesting.



02

Hands-on

Maps and tables

What you'll do

Hands-on

```
graph TD; Hands-on --- Geospatial; Hands-on --- Topical;
```

Geospatial

Pin points on a
map

Topical

Association rule
mining

Exercise

Assignment

Review

Review the tutorial

Exercise

Solve the problems
(time to code!)

Fill in the form
with your answers

TODO

Nothing, next time we
will work on web
technologies only

Thanks!

Do you have any questions?

marilena.daguino2@unibo.it

https://github.com/marilenadaquino/information_visualization

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